If the associated state contains a deferred function from a call to `std::async`, invokes that function and returns the result; otherwise, waits until the asynchronous result associated with an instance of `std::future` is ready, and then returns the stored value or throw the stored exception.

Declaration

```
void future<void>::get();
R& future<R&>::get();
R future<R>::get();
```

Preconditions

`this->valid()` would return `true`.

Effects

If the state associated with `*this` contains a deferred function, invokes the deferred function and returns the result or propagates any thrown exception.

Otherwise, blocks until the asynchronous result associated with `*this` is *ready*. If the result is a stored exception, throws that exception. Otherwise, returns the stored value.

Returns

If the associated state contains a deferred function, the result of the function invocation is returned. Otherwise, if `ResultType` is `void`, the call returns normally. If `ResultType` is `R&` for some type `R`, the stored reference is returned. Otherwise, the stored value is returned.

Throws

The exception thrown by the deferred exception or stored in the asynchronous result, if any.